

Modules

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1 Introduction to Module Theory

Let R, A be rings with identities $1_R, 1_A$, respectively. Suppose A is a left R -module such that for all $r \in R, a, b \in A$, we have

$$r \cdot ab = (r \cdot a)b = a(r \cdot b).$$

Let $\varphi : R \rightarrow A$ be given by $\varphi(r) = r \cdot 1_A$. Show that φ is a homomorphism defining an algebra structure of A over R and that the natural module structure of this algebra recovers the original module.

Suppose $r, s \in R$. Then

$$\varphi(r + s) = (r + s) \cdot 1_A = r \cdot 1_A + s \cdot 1_A = \varphi(r) + \varphi(s).$$

Further,

$$\varphi(r)\varphi(s) = (r \cdot 1_A)(s \cdot 1_A) = 1_A(r \cdot (s \cdot 1_A)) = rs \cdot 1_A = \varphi(rs).$$

Hence, φ is a ring homomorphism.

Further, for any $a \in A$, we have

$$(1_R \cdot 1_A)a = 1_A(1_R \cdot a) = 1_R \cdot a = a,$$

so $1_R \cdot 1_A = 1_A$, ie $\varphi(1_R) = 1_A$. Let $r \in R, a \in A$ be arbitrary.

$$\varphi(r)a = (r \cdot 1_A)a = 1_A(r \cdot a) = r \cdot a$$

and

$$a\varphi(r) = a(r \cdot 1_A) = (r \cdot a)1_A = r \cdot a,$$

so $\varphi(r) \in Z(A)$, so A is an algebra over R .

The natural action $\alpha : R \times A \rightarrow A$ is given by

$$\alpha(r, a) = \varphi(r)a = (r \cdot 1_A)a = 1_A(r \cdot a) = r \cdot a,$$

which equals the original module action.

2 Free Modules

Suppose F_0, F are free modules on A such that F_0 has the unique extension property. Then $F_0 \cong F$ and F also has the unique extension property.

Proof: Let ψ_0, ψ include A into F_0, F , respectively. By the extension property of F_0 , let $\phi : F_0 \rightarrow F$ be such that $\phi\psi_0 = \psi$. I claim that ϕ is a bijection.

ϕ is surjective. Let $f \in F$. Since F is free on A , F is generated by $\psi(A)$. There exist r_i, a_i such that $f = \sum r_i\psi(a_i)$. Then

$$f = \sum r_i\psi(a_i) = \sum r_i\phi\psi_0(a_i) = \sum \phi(r_i\psi_0(a_i)) = \phi\left(\sum r_i\psi_0(a_i)\right).$$

ϕ is injective. Suppose $\phi(f_0) = 0$ for some $f_0 \in F_0$. Since F_0 is generated by $\psi_0(A)$, $f_0 = \sum r_i\psi_0(a_i)$. Then

$$\phi(f_0) = \phi\left(\sum r_i\psi_0(a_i)\right) = \sum \phi(r_i\psi_0(a_i)) = \sum r_i\phi\psi_0(a_i) = \sum r_i\psi(a_i).$$

Since F is free on A , $(r_i) = 0$, so $f_0 = 0$, so ψ is bijective.

Suppose M is a left R -module and $\varphi : A \rightarrow M$ is a function. By extension property of F_0 , let $\Phi : F_0 \rightarrow M$ be such that $\Phi\psi_0 = \varphi$. Observe that $(\Phi\phi^{-1})\psi = \Phi\psi_0 = \varphi$, so there exists a map $\rho : F \rightarrow M$ such that $\rho\psi = \varphi$.

Suppose there exists another map $\rho' : F \rightarrow M$ such that $\rho'\psi = \varphi$. Then $\rho'\psi = \rho\psi$. Since $\psi(A)$ generates F , $\rho' = \rho$, so F has the unique extension property.

Fix a set A . A free module F_0 on A with the unique extension property exists. If F is any free module on A , then F has the unique extension property.

Now, if F, F' are two free modules with the unique extension property, there exists a unique isomorphism $\phi : F \rightarrow F'$ such that ϕ is constant over A .

Suppose $|A| = |B|$. Then $F(A) \cong F(B)$.

Let $h : A \rightarrow B$ be a bijection. Define $\Phi : F(A) \rightarrow F(B)$ by $f \mapsto fh^{-1}$. Φ is well-defined: suppose $b \in B(fh^{-1})$, so $fh^{-1}(b) \neq 0 \rightarrow h^{-1}(b) \in A(f) \rightarrow b \in h(A(f))$. Since $A(f)$ is finite, so is $h(A(f))$.

Suppose $f, g \in F(A)$. Then $\Phi(f+g) = (f+g)h^{-1} = fh^{-1} + gh^{-1} = \Phi(f) + \Phi(g)$. Further, for $r \in R$, $\Phi(rf) = (rf)h^{-1} = r(fh^{-1}) = r\Phi(f)$, so Φ is a module homomorphism.

Suppose $\Phi(f) = \Phi(g)$. Then $fh^{-1} = gh^{-1} \rightarrow f = g$. Suppose $f \in F(B)$. Then $\Phi(fh) = fhh^{-1} = f$. Indeed, if $a \in A(fh)$, then $fh(a) \neq 0 \rightarrow h(a) \in$

$B(f) \rightarrow a \in h^{-1}(B(f))$. Since $B(f)$ is finite, so is $h^{-1}(B(f))$.

Suppose R is commutative. If $R^n \cong R^m$, then $n = m$.

Let $I \triangleleft M$ be a maximal ideal. $R^n/IR^n \cong R^m/IR^m$. Note $R^n/IR^n \cong (R/IR)^n = (R/I)^n$. Since I is maximal, R/I is a field, so $\dim LHS = \dim RHS \rightarrow m = n$.

Let N be a submodule of M . Prove that if $M/N, N$ are finitely generated, then so is M .

Let $\{m_i + N\}_{i \in [u]}$ generate M/N . Let $\{n_j\}_{j \in [v]}$ generate N . I claim that $\{m_i\}_{i \in [u]} \cup \{n_j\}_{j \in [v]}$ generates M . Suppose $l \in M$. Then $l + N = \sum a_i(m_i + N) = \sum a_i m_i + N$, so $l - \sum a_i m_i \in N$. Then $l - \sum a_i m_i = \sum b_j n_j$, so $l = \sum a_i m_i + \sum b_j n_j$.

3 Irreducible Modules

A module M is said to be irreducible if $M \neq 0$ and the only submodules of M are $0, M$. Show that M is irreducible iff M is generated by any nonzero element.

Let $m \in M$ be nonzero. Then $Rm = M$ (since $m \in Rm$, so $Rm \neq 0$), so M is cyclic. Conversely, suppose $Rm = M$ for all $m \neq 0$. Let $N \neq 0$ be a submodule of M . Pick nonzero $n \in N$. Then $Rn = M$, so $Rn \subset N \subset M \rightarrow N = M$.

An irreducible $\mathbb{Z}$ -module is a simple nonzero Abelian group. Since M is cyclic, M is finitely generated. By the classification theorem, $M \cong \mathbb{Z}/p\mathbb{Z}$ for some prime p .

Let R be commutative. Suppose M is an irreducible R -module. Fix nonzero $m \in M$. Consider the map $\varphi : R \rightarrow M$ given by $r \mapsto rm$. For $r, s \in R$, we have $\varphi(r + s) = (r + s)m = rm + sm = \varphi(r) + \varphi(s)$. For $rs \in R$, we have $\varphi(rs) = (rs)m = r(sm) = r\varphi(s)$, so φ is a module homomorphism. Since $Rm = M$, φ is surjective.

Consider $I = \ker \varphi$ which is a submodule of R , ie an ideal. Note $I \neq R$ since $M \neq 0$. It suffices to show that R/I is a field. Suppose $a \notin I$, ie $am \neq 0$. Then $R(am) = M$, so for some $b \in R$ we have $b(am) = m$, or $(ab - 1)m = 0$, ie $ab - 1 \in I$. In other words, $(a + I)(b + I) = 1 + I$, as desired.

Suppose M_1, M_2 are irreducible R -modules. Let $\varphi : M_1 \rightarrow M_2$ be a nonzero module homomorphism. Then φ is an isomorphism.

Proof: Pick $m_1 \in M_1$ such that $m_2 = \varphi(m_1) \neq 0$. Then $m_1 \neq 0$. Apply

the previous result to m_1, m_2 , so $1 + I_1 \mapsto m_1$ and $1 + I_2 \mapsto m_2$ via the corresponding maps. We have a module homomorphism $\psi : R/I_1 \rightarrow R/I_2$ such that $\psi(1 + I_1) = 1 + I_2$. It follows that $\psi(r + I_1) = \psi(r(1 + I_1)) = r\psi(1 + I_1) = r(1 + I_2) = r + I_2$. Also,

$$\psi((r + I_1)(s + I_1)) = \psi(rs + I_1) = rs + I_2 = (r + I_2)(s + I_2) = \psi(r + I_1)\psi(s + I_1),$$

ie ψ is a field homomorphism. Observe ψ is a surjective field homomorphism sending $1 + I_1 \mapsto 1 + I_2$ and is therefore an isomorphism. Suppose $\varphi : M \rightarrow M$ is nonzero. Then φ is invertible, so $\varphi\varphi^{-1} = \varphi^{-1}\varphi = \text{id}_M$. Hence, $\text{End}_R(M)$ is a division ring (since elements need not be commutative).